

Maya Ravichandran

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EDUCATION

University of Oxford

Oxford, UK

MSc in Advanced Computer Science at New College

University of Cambridge

Cambridge, UK

MPhil in Therapeutic Sciences at Trinity College

Rutgers University–New Brunswick

New Brunswick, NJ

B.S. in Computer Science

AWARDS

Marshall Scholar: One of ~40 US citizens selected yearly by British government based on academic, leadership, & ambassadorial potential, receiving full funding for graduate studies in the UK at Oxford & Cambridge

Presidential Scholar: One of top 0.3% of applicants to Rutgers University-New Brunswick, receiving full scholarship for undergraduate studies

PUBLICATIONS

Ravichandran, M.*, Koch, M.*, Das, T.*, Khatri, N*. (2023). GraphRNN Revisited: An Ablation Study and Extensions for Directed Acyclic Graphs. **NeurIPS 2023: New Frontiers in Graph Learning Workshop**. ([Paper link](#))

WORK EXPERIENCE

Netflix

Fall 2024 - Present

Research Scientist

Los Gatos, CA

- ML researcher studying AI agents, focusing on research and production deployment of self-improving agent systems
- Designed and deployed a production agentic AI system supporting the release of 75+ Netflix movies/shows by autonomously analyzing internal communications and reconciling launch plans
- Built and led a 5-person research team investigating agent self-improvement and LLM tool use; manuscript in preparation
- Designed and ran comprehensive evals for production agentic systems, spanning both ML metrics and downstream business impact, using offline and online human annotation and automatic LLM-as-a-judge scoring
- Designing a data flywheel for agent self-improvement, enabling scalable learning from agent-user interactions
- Spearheaded adoption of agentic AI at Netflix by developing educational frameworks and training both technical and non-technical teams on integrating agentic AI into business systems
- Collaborating with central infrastructure team to set roadmap for a general software platform for AI agent deployment, evaluation, and continual learning
- Developed and released a system for ingesting key communications via a WebSocket server and actioning upon new messages via agentic AI models, built using Python, large language models, and AWS
- Promoted from ML Research Intern to Research Scientist based on impactful contributions

Regrello

Fall 2023 - Summer 2024

Machine Learning Engineer

San Francisco, CA

- Core contributor in the design and development of a multi-agent LLM system for generation of custom supply chain workflows, incorporating industry knowledge and customer-provided assets
- Incorporated AI-generated manufacturing processes into Regrello software to enable streamlined auto-execution of processes, using a backend written with Python and Flask, running on Docker, Kubernetes, and GCP
- Architected and deployed a scalable, distributed, asynchronous task queue using Celery in Python, enhancing the processing efficiency and reliability of LLM tasks in a high-demand environment
- Performed experiments to demonstrate the robustness of the multi-agent LLM system to prompt injection attacks
- Regrello was a Series A startup in the supply chain automation space. Regrello has been acquired by Salesforce and was funded by a16z.

Apollo Therapeutics

Spring 2023

Business Development Intern

Cambridge, UK

- Developed rapid screening method for identifying pharmaceutical drugs that would be strong acquisition candidates using filters on a pharmaceutical dataset, reducing number of drugs that needed to be manually examined by 77%

- Identified a viable acquisition candidate for Apollo

MongoDB

Summer 2021

Software Engineering (Machine Learning) Intern

New York, NY

- Developed a machine learning model for the novel application of predicting performance regressions based on code changes, using Python, Pandas, and Scikit-learn
- Achieved 0.88 accuracy and 0.91 ROC AUC score with passive-aggressive model, surpassing minimum accuracy requirement of 0.75
- Completed end-to-end machine learning development, including constructing a data pipeline integrating data from GitHub and a performance dataset, data preprocessing, feature engineering, and model prototyping and evaluation

MongoDB

Summer 2020

Software Engineering Intern

New York, NY

- Designed and implemented a data pipeline within MongoDB's distributed, open source continuous integration system
- Implemented pipeline in Go that logged system metrics from cloud hosts running test suites, streamed data to a data sink using gRPC, stored data using MongoDB and Amazon AWS S3, and made data accessible via REST API for diagnosis of system failures via machine learning and data visualization

Bank of America Merrill Lynch

Summer 2019

Sales and Trading Summer Analyst

New York, NY

- Designed and priced hedges using a custom basket of equities and an options collar
- Constructed five-year interest rate swap spreads to maximize revenue and minimize risk

Commvault

Summer 2018

Software Engineering Intern

Tinton Falls, NJ

- Designed and developed a data pipeline that collected user activity data and inputted it into ARIMA statistical prediction models using C++ for intelligent scheduling of background activities to enhance system availability for customers

Commvault

Summer 2017

Software Engineering Intern

Tinton Falls, NJ

- To improve CI/CD workflow for in-house software development by ~1,300 developers, created a full-stack application that contained a dynamic web interface using Angular, Bootstrap, HTML, CSS, Java, and MS SQL Server

TECHNICAL SKILLS

Languages: Python, Java, Go, C++, C, JavaScript, TypeScript, R, HTML, CSS, LaTeX

AI/ML: PyTorch, Pandas, Scikit-learn, LLMs, GPT-4, OpenAI API, Google Gemini

Tools/Frameworks: Angular, React, SQL, MongoDB, Flask

Developer Tools: Git, GitHub/GitLab, Docker, Kubernetes, GCP, AWS, Unix

RESEARCH EXPERIENCE

University of Oxford

Summer 2022

Machine Learning Researcher

Oxford, UK

- Trained natural language processing transformer models (based on BERT architecture, 110M parameters) and support vector machine (SVM) models on whole genome sequencing data to predict presence of Alzheimer's disease
- Using approach of SVM models applied to single nucleotide polymorphisms, achieved 0.65 ROC AUC, which was comparable to previous approaches using other methods

National Institutes of Health

Summer 2018

Bioinformatics Research Intern

Bethesda, MD

- Improved accuracy of probabilistic framework for discovery of structural variants (large-scale genome mutations) by eliminating false positives with machine learning, using R

Princeton University

Summer 2016 - Winter 2017

Materials Science Research Intern

Princeton, NJ

- Investigated the impact of sulfate attack on the atomic structure of eco-friendly, low-CO2 alkali-activated cement
- Identified changes to atomic bonds in cement using X-ray diffraction methods and X-ray pair distribution function analysis on data from Advanced Photon Source particle accelerator at Argonne National Laboratory
- Finalist at Intel International Science and Engineering Fair (ISEF), Los Angeles, 2017